WORKED SOLUTION

STEP II 1996, Question 6

Pure • Counting Divisors

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q6

The key idea

Every factor of N is built by choosing, independently for each prime p_i, how many copies of it to take, anywhere from 0 to m_i. Counting those choices counts all the factors, and then we throw away 1 and N. After that, both search questions become questions about exponents: which exponent patterns give the right count, and how to place those exponents on primes so that N is as small as possible.

Solution

The opening examples

The factors of 12 are $1, 2, 3, 4, 6, 12$. Removing 1 and 12 leaves $2, 3, 4, 6$, which is 4 proper factors. The factors of 16 are $1, 2, 4, 8, 16$. Removing 1 and 16 leaves $2, 4, 8$, which is 3 proper factors.

The general count

Suppose M divides N. Every prime factor of M is then a prime factor of N, and it cannot appear in M more often than in N. So

$$M = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}, \qquad 0 \le a_i \le m_i.$$

Conversely, every such product divides N. By uniqueness of prime factorisation, different choices of $(a_1, \ldots, a_r)$ give different integers M. So the factors of N correspond exactly to these choices. There are $m_i + 1$ choices for each a_i, made independently, so N has

$$d(N) = (m_1 + 1)(m_2 + 1) \cdots (m_r + 1)$$

factors in total. Since $r \ge 1$, we have $N > 1$, so 1 and N are two *different* factors. Removing them, the number of proper factors is

$$(m_1 + 1)(m_2 + 1) \cdots (m_r + 1) - 2.$$

Check: $12 = 2^2 \cdot 3$ gives $3 \cdot 2 - 2 = 4$, and $16 = 2^4$ gives $5 - 2 = 3$.

A principle for making N small

In both parts we fix the exponents and ask for the smallest N they allow. Two observations settle this.

- Use the smallest primes. If some prime q is used and a smaller prime p is not, replacing q by p keeps every exponent, so it keeps the count, and it makes N smaller. So the primes should be $2, 3, 5, \ldots$ with no gaps.
- Put the biggest exponents on the smallest primes. Suppose $p < q$, with p carrying exponent a and q carrying exponent b, where $a < b$. Swapping the two exponents multiplies N by $p^{b-a}/q^{b-a} < 1$, and it does not change the count.

So, for exponents $e_1 \geq e_2 \geq \cdots \geq e_r$, the smallest N is $2^{e_1}3^{e_2}5^{e_3} \cdots$.

Part (i)

We need exactly 12 proper factors, so $d(N) = 14$. Each factor $m_i + 1$ is at least 2, and $14 = 2 \cdot 7$ with 7 prime. So 14 can be written as a product of factors that are each at least 2 in only two ways:

- 14 itself: $N = p^{13}$, which is smallest at $2^{13} = 8192$;
- 7×2: $N = p^6 q$, which is smallest at $2^6 \cdot 3 = 192$.

So the smallest positive integer with exactly 12 proper factors is

$$\boxed{192 = 2^6 \cdot 3}.$$

Its factors are $1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, 96, 192$, and there are 14 of them.

Part (ii)

Now we need $d(N) \geq 14$. Since $d(N)$ is not fixed any more, we sort by the number r of distinct primes. Write the exponents in decreasing order, $e_1 \geq e_2 \geq \cdots$.

? **Ask yourself** Why can't we just reuse part (i) and look at $d(N) = 14$ only?

Because having *more* factors does not mean N has to be bigger. Exponents that are spread over several small primes give many factors cheaply. For example, $2^3 \cdot 3 \cdot 5$ has $4 \cdot 2 \cdot 2 = 16$ factors and is much smaller than 192. So we must look at every value $d(N) \geq 14$.

- $r \geq 4$: then $N \geq 2 \cdot 3 \cdot 5 \cdot 7 = 210$.
- $r = 3$: here $d(N) = (e_1 + 1)(e_2 + 1)(e_3 + 1) \geq 14$. If $e_2 = e_3 = 1$, we need $4(e_1 + 1) \geq 14$, so $e_1 \geq 3$ and $N \geq 2^3 \cdot 3 \cdot 5 = 120$. Otherwise $e_2 \geq 2$, so $e_1 \geq 2$ too, and $N \geq 2^2 \cdot 3^2 \cdot 5 = 180$.

- $r = 2$: here $(e_1 + 1)(e_2 + 1) \geq 14$ with $e_1 \geq e_2$. If $e_2 = 1$, then $e_1 \geq 6$ and $N \geq 2^6 \cdot 3 = 192$. If $e_2 = 2$, then $3(e_1 + 1) \geq 14$ gives $e_1 \geq 4$, so $N \geq 2^4 \cdot 3^2 = 144$. If $e_2 \geq 3$, then $e_1 \geq 3$ and $N \geq 2^3 \cdot 3^3 = 216$.
- $r = 1$: $e_1 + 1 \geq 14$, so $N \geq 2^{13} = 8192$.

Every case gives $N \geq 120$, and $N = 120 = 2^3 \cdot 3 \cdot 5$ actually works: it has $4 \cdot 2 \cdot 2 - 2 = 14$ proper factors. So the smallest positive integer with at least 12 proper factors is

$$\boxed{120},$$

which in fact has 14 proper factors.

Checks

A brute-force count of proper factors for every N from 2 to 9000 agrees with the formula $d(N) - 2$. It also confirms that 192 is the first N with exactly 12 proper factors, and that 120 is the first with at least 12 (it has 14). For a quick hand check, the numbers just below 120 with lots of factors, namely $60, 72, 84, 90, 96$ and 108, all have 12 factors and so only 10 proper ones.

Common slips

- Setting $d(N) = 12$ instead of 14. Remember that 1 and N are excluded.
- Answering 192 in part (ii) too. The smallest N with at least 12 proper factors need not have exactly 12.
- Putting the big exponent on the big prime: $2 \cdot 3^6$ has the same count as $2^6 \cdot 3$ but is far larger.

Beyond STEP

The divisor function $d(N)$ is *multiplicative*: $d(ab) = d(a)d(b)$ whenever a and b are coprime. That is exactly what the product formula expresses. Numbers with more factors than every smaller number are called *highly composite*, and Ramanujan studied them in depth. 120 is one of them, and the exponent-ordering argument used here is the starting point of his analysis. Davenport's *The Higher Arithmetic* is a lovely introduction to divisor functions and unique factorisation.

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